Mid-semester Exam Practice Questions

EGH445 Modern Control

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Instructions: This quiz has a total of 3 problems, with several sub-problems.
The total number of possible marks for this quiz is 100. It is recommended that you attempt to answer all questions. All answers need to be justified. Explanations of the steps used are essential and will be carefully accounted for in the assessment. Best marks are awarded to numerically correct solutions, but partial credits may be given to partially complete solutions. Any assumptions made should be consistent with the question, clearly stated and justified.

Problem 1

Consider the system

$$\ddot{y}(t) + y(t) = -0.5\dot{y}(t) + u(t), \tag{1}$$

where $u(t)$ is the input and $y(t)$ is the (scalar) output.

A) Convert (1) into state-space form.

B) Is the system (1) linear and time-invariant? (Justify your answer).

C) Is the system (1) completely controllable? (Justify your answer).

Problem 2

Consider the linear system

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned} \tag{1}$$

A) Is the system internally stable? Is it BIBO stable? (Justify your answers).

B) Now suppose that a different actuator has been chosen so that

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u. \end{aligned} \tag{2}$$

Is the system internally stable? Is it BIBO stable? How do your findings compare to part A) above? (Justify your answers)

Problem 3

Consider a reaction occurring during the extraction of aluminium, modelled in state-space form as follows:

$$\dot{x} = \phi(x, u), \quad y = \eta(x, u) \tag{1}$$

where $x = [x_1\, x_2]^T$ and

$$\phi(x, u) = \begin{bmatrix} -x_1 x_2 \\ -x_2^2 - x_1 x_2 + u \end{bmatrix}, \quad \eta(x, u) = x_2.$$

In the above, $x_1(t) > 0$ and $x_2(t) > 0$ denote non-metal and metal quantities in the reaction chamber, whereas the scalar input $u(t)$ is used to control the reaction temperature. The quantity $x_2(t)$ is measured during the reaction.

A) Determine all α such that

$$\bar{x}_\alpha = \begin{bmatrix} 0 \\ \alpha \end{bmatrix} \tag{2}$$

is an equilibrium point of the model when $\bar{u} = 4$ and interpret from a practical viewpoint.

B) Write the linearised state-space model, in matrix form, about the equilibrium point(s) identified above in part A). Is this model time-invariant?

C) Determine the stability of the equilibrium point(s) identified above in part A).

Problem 4

Consider the linear system

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} -3 & 0 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} u \\ y &= \begin{bmatrix} 1 & \epsilon \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \end{aligned} \tag{3}$$

where ϵ, c_1 and c_2 are scalar parameters.

A) Compute the system transfer function $G(s)$.

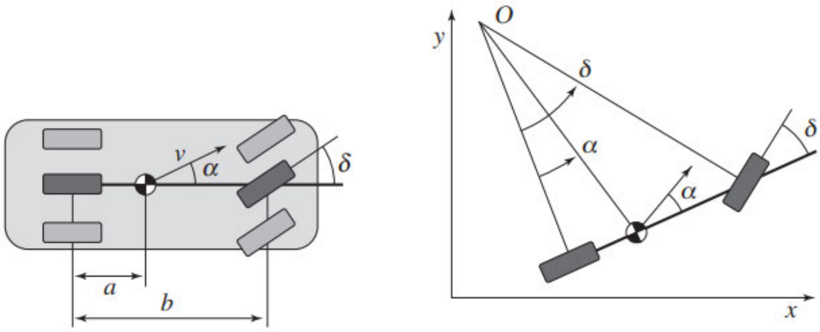
B) For which parameter values ϵ, c_1 and c_2 is the system not completely observable?

C) Interpret the cases identified in point B) above, taking into account the order of $G(s)$ and of the model in (3).

(cont/...)

Problem 5

Consider the vehicle steering system and its simplified “bicycle model”, as below



(a) Overhead view (b) Bicycle model

The nonlinear equations of motion for the system can be written as

$$\frac{d}{dt}\begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} v\cos(\alpha(\delta)+\theta) \\ v\sin(\alpha(\delta)+\theta) \\ \dfrac{v\sin(\alpha(\delta))}{a} \end{bmatrix}, \quad \alpha(\delta) = \arctan\left(\frac{a\tan\delta}{b}\right), \tag{1}$$

where the controlled input is the steering angle δ. Suppose that we are concerned only with the lateral deviation of the vehicle from a straight line. This corresponds to the equilibrium point $\bar{\theta} = 0$ and driving along the x axis with constant velocity v_0. Further consider $b = 3a$ and make use of the truncated Taylor series expansion

$$\arctan\left(\frac{a\tan\delta}{b}\right) \approx \frac{\delta}{3} + \frac{\delta^3}{10}.$$

We can then focus on the equations of motion in the y and θ directions and describe the system by:

$$\dot{z} = \begin{bmatrix} v_0\sin(\alpha(\delta)+z_2) \\ \dfrac{v_0\sin(\alpha(\delta))}{a} \end{bmatrix}, \quad y = z_1, \tag{2}$$

A) Find an equilibrium point and linearise the model (2) about it.

B) Investigate whether the equilibrium point considered above is stable.

C) Is the linearised model controllable?

D) In plain words, what does it mean for this system to be controllable?

Problem 6

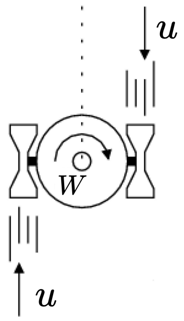


Figure 1: Monoaxial satellite.

Consider a monoaxial satellite rotating around a fixed inertial frame in weightless space. The satellite is driven by torques induced by the controlled expulsion of gases acting in opposite directions, as illustrated in Figure 1. A simplified nonlinear satellite model is given by the following state-space equations:

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(1 + x_1(t)^2)x_2(t) \\ \frac{1}{W}u(t) \end{bmatrix}, \tag{1}$$

where $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$ denote, respectively, the angular position and velocity of the satellite with respect to the inertial frame, $u \in \mathbb{R}$ is the control input (the effective applied torque to the satellite), and W is its moment of inertia with respect to its center of mass. Assume that $W = 1$.

A) Find all equilibrium points and linearise the model (1) around them.

B) Investigate whether the equilibrium points considered above are stable.

C) Are the linearised models controllable?

Problem 7

Consider the linear time-invariant system

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u(t), \tag{2}$$

with states $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$, and control input $u \in \mathbb{R}$.

A) As engineers, we seek to minimize costs and use as few sensors as possible to operate the system. We have two options: 1) install a sensor that measures $x_1(t)$; or 2) install a sensor that measures the quantity $x_1(t)+x_2(t)$. Which of these two sensors should we select such that, by using this one sensor, the complete state $(x_1(t), x_2(t))$ can be estimated asymptotically?